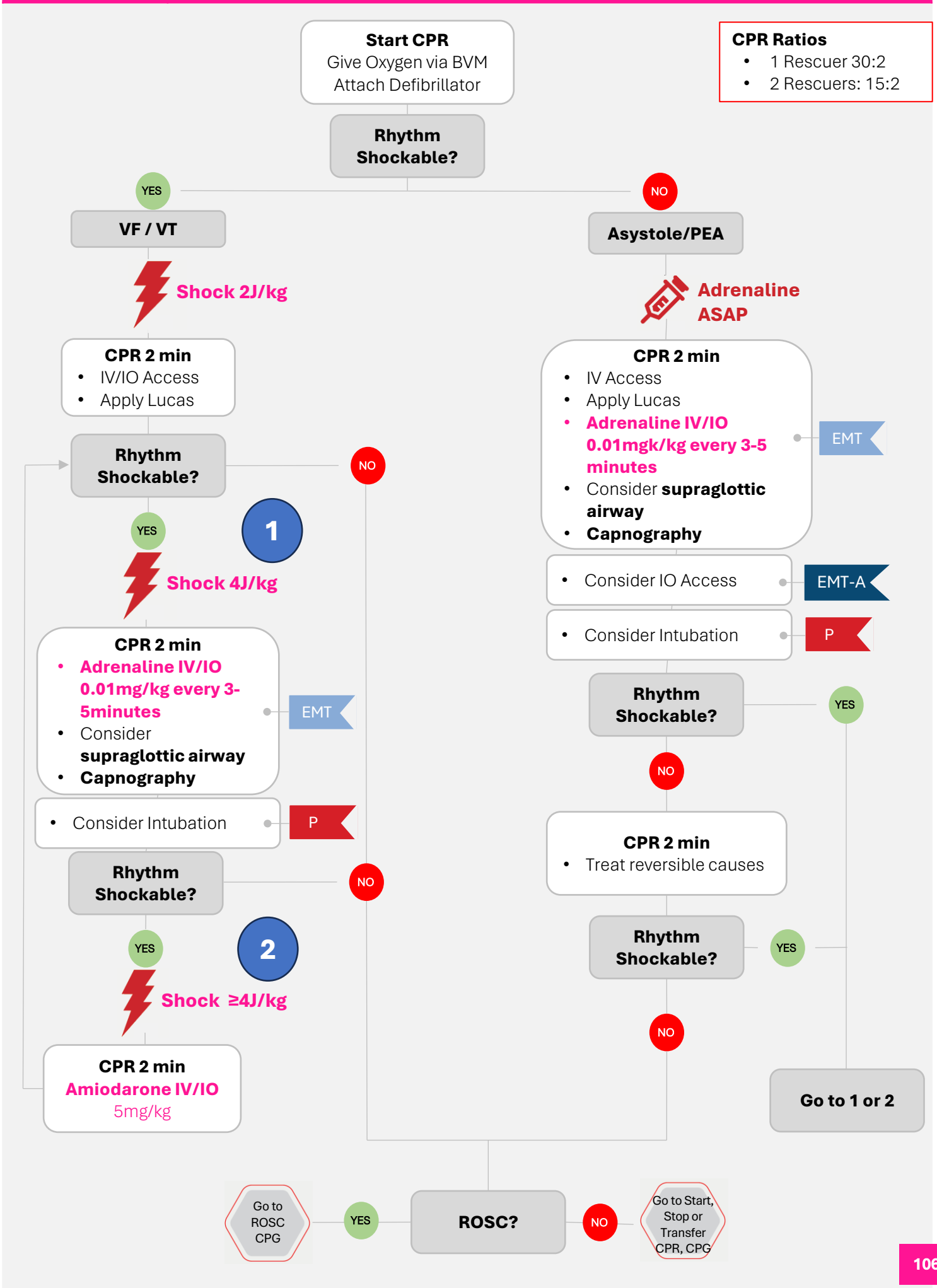
Pediatric Cardiac Arrest (<13 years)



Patient Care Goals

1. Recognize pediatric cardiac arrest immediately and begin resuscitation without delay.
2. Deliver high-quality CPR with proper rate, depth, and minimal interruptions to optimize coronary and cerebral perfusion.
3. Provide effective ventilations and oxygenation to correct hypoxia and prevent secondary injury.
4. Achieve early defibrillation if a shockable rhythm is present to improve chances of ROSC.
5. Identify and correct reversible causes of arrest (4 Hs & 4 Ts) to increase likelihood of successful resuscitation.
6. Minimize hands-off time during CPR and prioritize continuous compressions.
7. Ensure clear team communication and role assignment to maximize efficiency and reduce errors during resuscitation.
8. Prevent hyperthermia and hypoglycemia, which can worsen neurological outcomes.
9. Safely stabilize and prepare the child for transport, ensuring seamless handover to the receiving facility.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Infant or child ≤13 years, unresponsive, not breathing normally, and pulseless.

Exclusion Criteria

- Adult patients (>13 years)

Key Patient Assessment Points

1. **Scene Safety & Initial Impression**
   - Ensure the scene is safe before entering.
   - Rapid visual assessment/General Impression of the child (Pediatric Assessment Triangle – PAT):
     - Appearance: Unresponsive, no purposeful movement, limp.
     - Work of Breathing: Apneic or agonal gasps (not effective breathing).
     - Circulation to Skin: Pale, mottled, cyanotic.
2. **Confirm Unresponsiveness**
   - Tap the child and shout their name.
   - Look for any purposeful movement or eye opening.
3. **Airway & Breathing**
   - Open airway using head-tilt-chin-lift (if no trauma) or jaw-thrust (if trauma suspected).
   - Check for normal breathing for no more than 10 seconds.
   - Look for chest rise and listen for breath sounds; agonal breaths or gasping are not normal breathing.
4. **Circulation**
   - Check for a central pulse (carotid in children >1 year; brachial in infants <1 year) for no more than 10 seconds.
   - Assess skin color, temperature, and capillary refill time.
   - If no pulse or HR <60 bpm with poor perfusion despite oxygenation & ventilation → start CPR.
5. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - All patients cardiac arrest are Red Priority.
6. **Determine Likely Etiology of Cardiac Arrest (where possible)**
   - Gather information from caregivers/bystanders:
     - Onset & events leading to arrest (e.g., choking, drowning, trauma, sudden collapse).
     - Known medical conditions (e.g., congenital heart disease, asthma).
     - Medications, toxins, or recent illnesses (fever, diarrhea, seizures).
7. **Check for Reversible Causes** (4 Hs & 4 Ts)
   - Hypoxia, hypovolemia, hypothermia, hypo-/hyperkalemia.
   - Tamponade, tension pneumothorax, toxins, thrombosis.
8. **Assess for Trauma or Environmental Factors**
   - Inspect for signs of trauma, bleeding, burns, or exposure to cold/heat.

Patient Management & Interventions

1. **Request ALS Back-Up**
   - Request for ALS back-up as per Universal Care CPG if not already dispatched.
   - Specify "pediatric cardiac arrest" and confirm ETA of ALS.
2. **Start CPR Immediately:**
   - **Compressions:**
     - Rate: 100–120/min.
     - Depth: ≥1/3 chest depth (~4 cm infants, ~5 cm children).
     - Allow full chest recoil; minimize interruptions (<10 seconds).
   - **Compression: Ventilation Ratios:**
     - Single rescuer: 30:2.
     - Two or more rescuers: 15:2.
   - Rotate compressors every 2 minutes if possible.
3. **Airway & Ventilation:**
   - Open airway using head-tilt-chin-lift (Use Jaw thrust if trauma is suspected).
     - Use OPA/NPA when available.
     - Supraglottic Airways (Laryngeal Tubes) should be routinely placed when available.
   - Provide 2 breaths after each set of compressions.
   - Transition to BVM with 100% oxygen as soon as available.
   - Target SpO_2 ≥94%, avoid hyperoxia.
   - Maintain $EtCO_2$ ~35–45 mmHg if monitored.
   - **Consider Advanced Airway Management:**
     - See Airway Management CPG.
     - Intubation:
       - Indications:
         - If the Supraglottic Airway is not providing adequate ventilations, or/
         - If there is excessive secretions and/or vomit in the airway.
       - ETT placement should always be confirmed with waveform capnography (ETCO2).
4. **Defibrillation**
   - Attach AED or manual defibrillator as soon as it is available.
   - If pediatric pads are not available, use adult pads.
   - If shockable rhythm (VF/pVT):
     - First shock: 2 J/kg.
     - Subsequent shocks: 4 J/kg, up to max 10 J/kg.
   - Resume CPR immediately after shock.
5. **Obtain IV Access**
   - If IV access is not readily available or may be delayed, consider IO access.
6. **Administer Epinephrine (Adrenaline):**
   - Dose: 0.01 mg/kg (1:10,000) IV/IO every 3–5 min.
   - If no IV/IO access is available, consider administration via ETT at 0.03mg/kg.
7. **Administer Amiodarone**
   - If refractory VF/pVT:
     - After 3rd Shock: 5mg/kg, IV/IO bolus.
     - After 5th Shock: 5mg/kg, IV/IO bolus.
8. **Consider Reversible Causes**
   - Check Blood Sugal Level. If hypoglycaemia discovered, administer treatment as per Hypoglycaemia CPG.
   - Assess & correct 4 Hs & 4 Ts:
     - Hypoxia, hypovolemia, hypothermia, hypo-/hyperkalemia
     - Tension pneumothorax, tamponade, toxins, thrombosis.
9. **Reassess**
   - Check for pulse (ROSC) every 2 minutes.

10. Transfer

- Pediatric patients in cardiac arrest must be transferred to hospital.
- See Starting, Stopping or Transferring CPR CPG for further guidance when transferring under CPR.

Patient Safety Considerations

- Avoid over-ventilation or excessive oxygenation.
- Use correct compression depth and rate.
- Defibrillate promptly if indicated; avoid delays.
- Rotate compressors to prevent fatigue.
- Protect airway & spine if trauma suspected.
- If ALS is delayed but hospital is closer, consider transporting with ongoing CPR if it is safe to do so.

Clinical Insights & Additional Guidance

- Cause of Cardiac Arrest
  - Pediatric cardiac arrest is often secondary to hypoxia — prioritize effective ventilation and oxygenation.
- Airway Management
  - Airway management in pediatrics may be difficult.
  - Supraglottic (LT/laryngeal tube) airways should be routinely used and are generally preferred over OPA's.
  - Intubation in pediatrics is difficult, and the chances of first attempt success is likely lower than adults.
    - Careful airway assessment and team coordination is required to ensure that intubation is conducted in the appropriate patient and interruptions to ventilation and compressions are minimized.
- Defibrillation using Lifepak 15
  - Using Pediatric Pads with the Lifepak 15 in AED mode does not adjust the Joules appropriately for pediatric patients – manual mode must be used, and the appropriate Jules selected.
  - Defibrillation should be started a 2 J/kg then escalated to 4J/kg then escalated to ≥4J/kg
    - Joules should not exceed 10J/kg or the adult dose.
- Equipment
  - Always use appropriate-sized equipment for infants and children; the Broselow tape is recommended to quickly estimate weight and select correct equipment sizes and medication doses.
- Team Coordination
  - Communicate clearly and assign roles during resuscitation.
- Advanced Life Support Availability
  - If ALS backup unavailable, maintain effective CPR & transport if hospital is closer.

Documentation

- Identification: Name, age, sex, weight (if known).
- Medical complaint: Pediatric cardiac arrest (witnessed or unwitnessed).
- Information/Injuries: Suspected cause, relevant history (trauma, drowning, medical).
- Signs: Initial and ongoing HR, rhythm, SpO_2, $EtCO_2$, responsiveness.
- Treatment: CPR start time, quality (rate/depth), number & energy of shocks, medications, airway interventions.
- Allergies: If known.
- Medications: Drugs given during resuscitation.
- Background & Other: Any medical history, bystander CPR, time of collapse, name & role of receiving staff.

Key Performance Indicators

1. Initiate CPR within 1 minute of arrest recognition.

References

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